Forecasting Daily Traffic Incidents on Pasig City's Major Roads – Case Study Documentation

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Course: IT 201 – Platform Technologies

Chapter 1 – Problem Identification and Data Requirements

Pasig City's three busiest roads (C5, Ortigas Avenue and Marcos Highway) averaged 0.87 MMDA-reported traffic incidents per road per day between August 2018 and February 2020, with C5 alone averaging 1.64. Traffic managers have no day-ahead estimate of how many incidents to expect on each road, so this project builds one.

1.1 The problem

Vehicular accidents, collisions, stalled vehicles and road works block lanes and slow traffic on Pasig's major roads. Enforcers and tow trucks are placed where incidents are expected, but the expected number is a judgement call, not a forecast. The project falls under the case study's suggested problem area "Pasig City traffic conditions".

To validate the problem, the group reviewed published reports (preliminary research). MMDA's Metro Manila Accident Recording and Analysis System (MMARAS) recorded 121,771 road crashes in Metro Manila in 2019, about 334 a day, with 394 deaths (Top Gear Philippines, 2020). The same MMARAS data showed C5 as Metro Manila's deadliest road for the third year running, with 31 fatalities, 1,136 non-fatal injuries and 7,568 property-damage cases in 2019 (Chua, 2020). The project's own dataset shows that MMDA reported 1,611 incidents on C5, Ortigas Avenue and Marcos Highway within Pasig City over about 18 months (August 2018 to February 2020). MMARAS counts crashes while the dataset counts incidents that MMDA tweeted, so the two are not the same measure, but both show that incidents on these roads are frequent and well documented. News reports on MMDA data also show that most Metro Manila road crashes in 2024 took place on EDSA and C5 (Sigales, 2025), and that traffic cost the Philippines about P3.5 billion per day in 2018 (Wheels.ph, 2019). Because enforcers and tow trucks are limited, knowing how many incidents to expect on each road and day is a practical planning question.

1.2 Who is affected and why it matters

Commuters, motorists and public transport riders lose time when an incident blocks a lane, and they face risk at the incident site. Residents and businesses along the roads are affected by the congestion that incidents cause. MMDA and Pasig City traffic management personnel decide where to send limited enforcers and tow trucks, so they are the direct users of any forecast.

Relevance to Pasig City and PLP: C5, Ortigas Avenue and Marcos Highway are major roads that serve Pasig City, and 1,723 of the dataset's 17,312 rows (10%) are incidents that MMDA placed in Pasig City. The results are meant for the Pasig City traffic management office and the MMDA personnel who cover Pasig, and they also matter to Pasig commuters, including PLP students and staff who travel on these roads.

The data shows the pressure is uneven: half of all road-days have no reported incident, while C5 regularly has two or more. Knowing which road and which days are likely to be busy lets scarce units go where they are needed.

1.3 Objectives and research questions

General objective: To develop and evaluate a regression model that predicts the number of MMDA-reported incidents per day on each of three major roads in Pasig City.

Specific objectives

The specific objectives are to prepare a clean road-day dataset from public MMDA traffic alerts; to explore how incidents vary by road, day of week and hour; to train regression models and compare them against a simple per-road average; and to assess whether the best model is useful for planning and state its limits.

Research questions

The study asks four research questions.

1. How do reported incident counts differ by major road and by day of week?
2. Does recent activity, measured over the previous 7 days, help predict today's count?
3. Does a regression model predict daily counts more accurately than a per-road average?
4. Is the model accurate enough to support planning, and where does it fall short?

1.4 Data needed

Data	Used for
Date of each incident	Build daily counts, day of week, month and recent-activity features
Location name	Assign each incident to C5, Ortigas Avenue or Marcos Highway
Time	Hour-of-day analysis
Incident type, lanes blocked, vehicles involved	Describing the incidents (not used by the model)
Tweet URL	Tracing each record to its original alert

The data must cover enough days to learn weekly patterns and to test on later days than those used for training.

1.5 Machine learning task and algorithm

Task: Regression. The target is the number of incidents on a road in a day (0, 1, 2, ...).

Algorithms: Linear Regression, Poisson Regression and Random Forest Regressor, each compared with two baselines that predict every road's average (mean) or median count. Chapter 5 explains why each is appropriate.

1.6 How the results could be used

The predicted counts could help the MMDA and Pasig City's traffic management office plan the weekly placement of enforcers and tow trucks, for example by sending more units to the roads and days expected to be busiest. The model is a planning aid, not an automated decision-maker; Chapter 7 discusses its limits.

Chapter 2 – Data Gathering

The project uses a public dataset of MMDA traffic alerts for Metro Manila (17,312 rows, 20 Aug 2018 to 27 Dec 2020), of which 1,723 rows are for Pasig City. The original file was kept unchanged as the raw dataset.

2.1 Source and method

The project uses a publicly available dataset with a documented source, which is one of the options Phase II allows. The dataset is *Manila Traffic Incident Data: Incidents Reported by the Metro Manila Development Authority Twitter Page*, shared on Kaggle by Panji Brotoisworo under the CC BY-NC-SA 4.0 license (https://www.kaggle.com/datasets/esparko/mmda-traffic-incident-data). The file used is `data_mmda_traffic_spatial.csv`.

The original source is the MMDA ALERT tweets on the MMDA Twitter page. According to the dataset's description, the author collected them through the Twitter API and parsed the tweet text with a Python script (https://github.com/pbrotosworo/tweet2map). The data starts on 20 August 2018 because of the author's free-tier API access, and there is a gap in early 2020 caused by pandemic-related difficulties and a code upgrade. The author notes that the dataset does not represent all incidents in Metro Manila: the MMDA's jurisdiction is mainly the national roads, so the data is biased toward roads such as EDSA and C5. The author is an independent researcher with no affiliation with the MMDA. Every record links to its tweet in the `Source` column.

2.2 Raw dataset preservation

The raw file is stored unchanged. All cleaning is done in memory in the notebook, so the original can always be re-checked.

2.3 Structure of the raw data

The file has 17,312 rows and 13 columns.

Column	Type	Meaning
Date	Text (YYYY-MM-DD)	Date of the alert
Time	Text	Time of the alert, for example "7:55 AM"
City	Text	City where MMDA placed the incident
Location	Text	Road segment or landmark named in the alert
Latitude, Longitude	Decimal number	Geocoded position of the location
High_Accuracy	0 or 1	1 if the geocoding is precise
Direction	Text	NB, SB, EB or WB
Type	Text	Incident type, for example vehicular accident or stalled vehicle
Lanes_Blocked	Number	Number of lanes occupied
Involved	Text	Vehicles involved
Tweet	Text	Alert text
Source	Text (URL)	Link to the original tweet

2.4 Scope and sufficiency

Stage	Rows
Raw file, all of Metro Manila	17,312
Rows where City is Pasig City	1,723 (10%)
Incidents on the three major roads, after cleaning	1,611
Road-day rows used for modeling	1,656

2.5 Data quality at collection

In the 1,723 Pasig rows, `Time` is missing in 8 rows, `Direction` in 46, `Type` in 5, `Lanes Blocked` in 49 and `Involved` in 40. Across the whole file, 187 rows have no `City`. There are 242 different spellings of location and 111 incident type labels, so they must be grouped.

2.6 Privacy and consent

The records are public posts by a government agency. They contain no names, plate numbers or other personal details, only vehicle types and places, so no individual consent is needed. The source is cited, and the dataset's license (CC BY-NC-SA 4.0) allows non-commercial use with attribution. Any derived dataset that is shared should carry the same license. No personal or sensitive data was added.

Chapter 3 – Data Preparation

The 17,312 raw rows became an engineered dataset of 1,656 rows and 8 columns, one row per major road per day, saved as `engineered_pasig_major_road_day.csv`. Each step below says what was done, why, and how many rows remained.

3.1 Steps, treatment and justification

Step	What was done	Why	Rows after
1. Filter	Kept rows where City is Pasig City	The study is about Pasig	1,723
2. Dates	Converted `Date` from text to a date type	Needed for day-of-week, month and lag features	1,723
3. Hour	Read the hour from `Time`; where unusable (10 rows), recovered it from the alert text in 3 rows	Hour is only used for one exploratory chart, so the 7 unrecoverable rows were kept	1,723
4. Duplicates	Removed 3 repeated tweets (same URL) and 35 cases of the same incident tweeted twice (same date, time, location, direction and type)	Avoid counting one incident twice	1,685
5. Major roads	Grouped 242 location spellings into C5, Ortigas and Marcos Highway; excluded 74 rows (EDSA, Meralco Onyx, ADB, Poveda and others)	Daily counts need enough incidents per group; the other locations are too scattered	1,611
6. Validation	Checked ranges and categories (see	Catch invalid values	1,611

Step	What was done	Why	Rows after
	3.2)		
7. Period	Kept dates up to 29 Feb 2020	Counts fall sharply from March 2020 (about 90 incidents in Feb 2020, 29 in Mar, 1 in Apr); the dataset's author reports a collection gap in early 2020, and traffic also fell during the lockdown	–
8. Aggregate	Counted incidents per major road per day and added zero-incident days	The file lists only days with at least one incident	1,677
9. Features	Added day of week, month, weekend flag, the count 7 days earlier and the average of the previous 7 days	Capture weekly and recent-activity patterns	1,677
10. Lag rows	Dropped the first 7 days, which have no history	Lag features need 7 earlier days	1,656

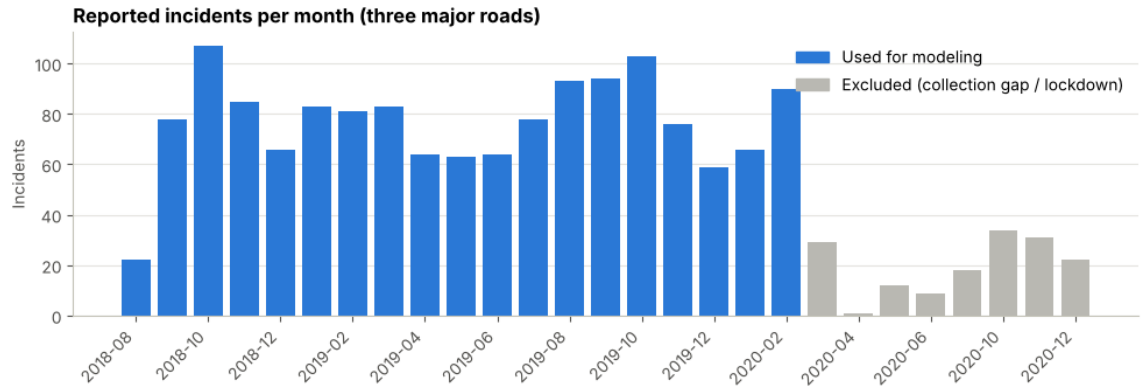


Figure 3.1 – Reported incidents per month. Months after February 2020 (grey) are excluded from modeling.

3.2 Validation of ranges, units, dates and categories

All checks on the 1,611 incidents passed: latitude within 14.50 to 14.66 and longitude within 121.03 to 121.13 (the Pasig area); `High_Accuracy` is 0 or 1; `Direction` is NB, SB, EB or WB when present; `Lanes_Blocked` is between 1 and 6 when present; hours are between 0 and 23; dates fall between Aug 2018 and Dec 2020. Column names were already consistent, so no renaming was needed.

3.3 Missing values

Hour (7 rows), Direction (37), Type (5), Lanes_Blocked (42) and Involved (39) are missing in the 1,611 incidents. These are descriptive columns that the model does not use, so they were not imputed and the rows were kept.

Days with no row in the file are treated as days with no reported incident. This is an assumption, because a day could also be missing when MMDA posted nothing, and the data cannot tell the two apart.

3.4 Outliers

The highest counts are 7 incidents (C5 on 16 Jan 2019 and 19 Feb 2019) and 6 incidents on 7 other days, all on C5. They are plausible busy days, not errors, so they were kept. Across all road-days the mean is 0.87 and the maximum is 7.

3.5 Derived variables

Variable	Meaning	Used by the model
DayOfWeek	0 = Monday to 6 = Sunday	Yes
Month	1 to 12	Yes
IsWeekend	1 on Saturday or Sunday	No (duplicates DayOfWeek; saved for reference)
Incidents_Lag7	Count on the same road 7 days earlier	Yes
Incidents_Roll7	Average count on the same road over the previous 7 days, excluding today	Yes

3.6 Encoding and scaling

Major road (3 values), day of week (7) and month (12) are one-hot encoded into 22 columns. The two lag features are standardized (mean 0, standard deviation 1). Together with the 2 scaled features this gives 24 model inputs. Both steps are fitted on the training period only and are re-fitted inside every cross-validation fold, so nothing from the test period leaks into training.

3.7 Resulting dataset

Dataset	Rows	Columns
Raw file	17,312	13
Pasig City rows	1,723	13
Engineered dataset	1,656	8

The engineered columns are `Date`, `MajorRoad`, `DayOfWeek`, `Month`, `IsWeekend`, `Incidents_Lag7`, `Incidents_Roll7` and `Incidents` (the target). It covers 27 Aug 2018 to 29 Feb 2020: 552 days for each of the 3 roads, with no missing values.

Chapter 4 – Exploratory Data Analysis

Half of all road-days (843 of 1,656) have no reported incident, C5 averages 1.64 incidents a day against 0.54 and 0.43 on the other two roads, and weekdays are busier than weekends. These patterns are the basis for the model's features.

4.1 First and last five rows of the engineered dataset

Date	MajorRoad	DayOfWeek	Month	IsWeekend	Incidents_Lag7	Incidents_Roll7	Incidents
2018-08-27	C5	0	8	0	2	0.857	0

Date	MajorRoad	DayOfWeek	Month	IsWeekend	Incidents_Lag7	Incidents_Roll7	Incidents
2018-08-27	MARCOS	0	8	0	0	0.286	1
2018-08-27	ORTIGAS	0	8	0	1	0.429	1
2018-08-28	C5	1	8	0	0	0.571	3
2018-08-28	MARCOS	1	8	0	1	0.429	0
...	...	...	...	...	...	...	...
2020-02-28	MARCOS	4	2	0	0	0.143	1
2020-02-28	ORTIGAS	4	2	0	0	0.000	2
2020-02-29	C5	5	2	1	0	1.857	2
2020-02-29	MARCOS	5	2	1	0	0.286	1
2020-02-29	ORTIGAS	5	2	1	0	0.286	1

4.2 Descriptive statistics

Variable	Mean	Std	Min	25%	50%	75%	Max
Incidents (target)	0.872	1.170	0	0	0	1	7
Incidents_Lag7	0.866	1.167	0	0	0	1	7
Incidents_Roll7	0.866	0.702	0	0.429	0.571	1.143	3.429

4.3 Distribution of the target and frequency of categories

Incidents in a road-day	Road-days	Share
0	843	50.9%
1	436	26.3%
2	229	13.8%
3	78	4.7%
4 or more	70	4.2%

The target is right-skewed: most road-days have 0 or 1 incident and a few have up to 7. Each of the three roads has 552 road-days, so the categories are balanced by construction.

4.4 Visualizations

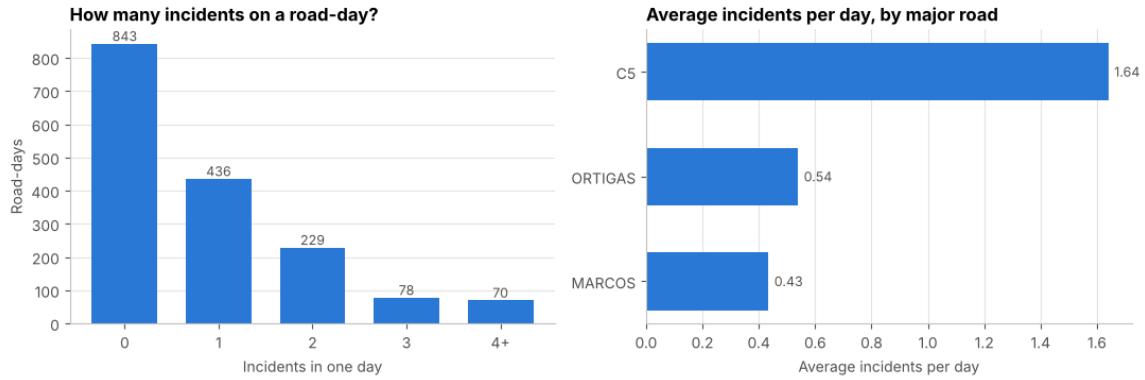


Figure 4.1 – Distribution of daily incidents (left) and average incidents per day by major road (right).

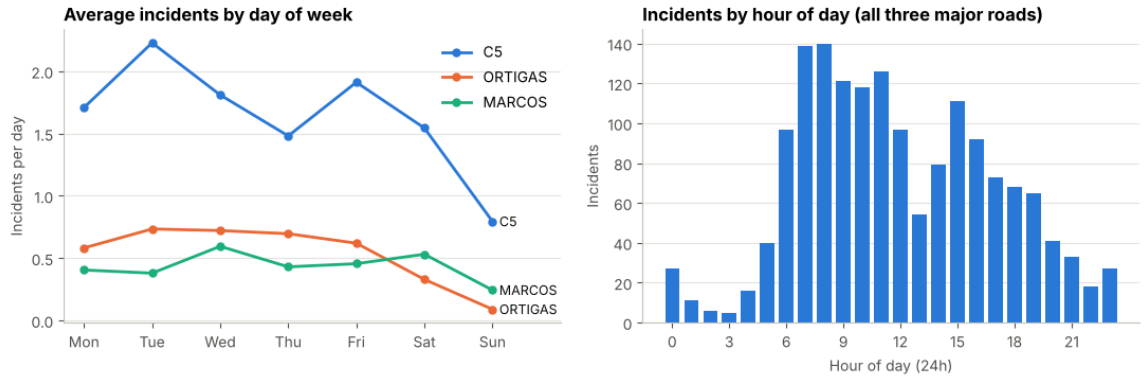


Figure 4.2 – Day-of-week pattern for each road (left) and incidents by hour of day across the three roads (right).

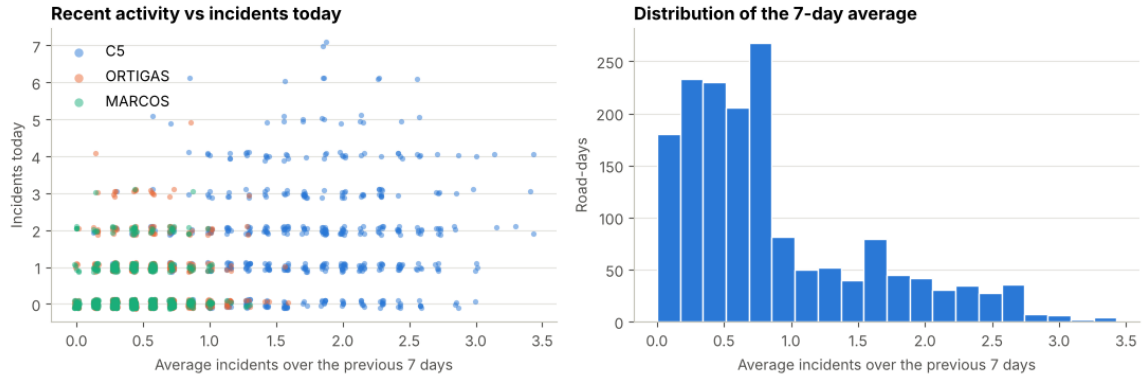


Figure 4.3 – Recent activity against incidents today, by road (left) and distribution of the 7-day average (right).

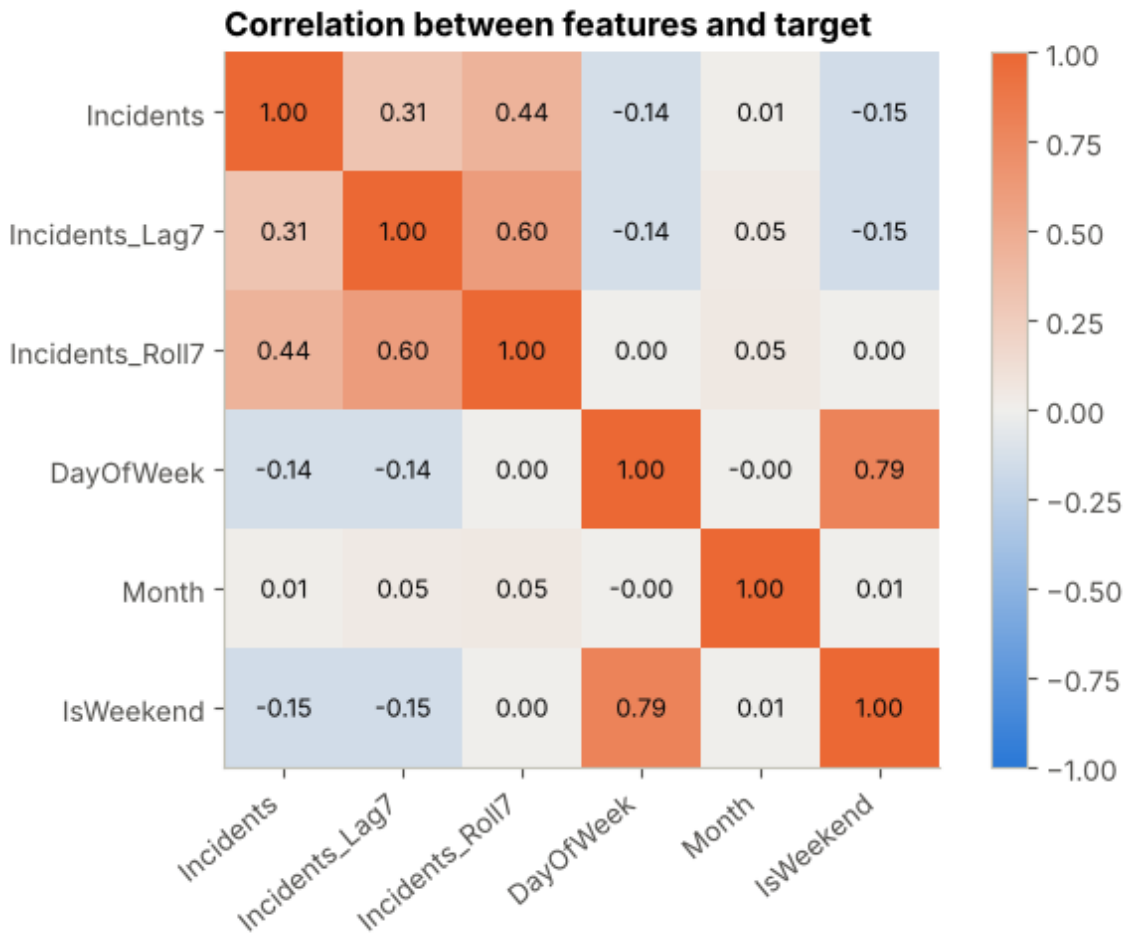


Figure 4.4 – Correlation between the numeric features and the target.

4.5 Findings and their relevance

The major road matters most. C5 averages 1.64 incidents per day, against 0.54 on Ortigas and 0.43 on Marcos Highway. This is relevant because the road is the strongest predictor, and the per-road average is the natural baseline to beat.

Weekdays are busier than weekends. C5 peaks on Tuesdays (about 2.2) and falls to about 0.8 on Sundays, while Ortigas drops to almost zero on Sundays. Day of week correlates weakly and negatively with the target (-0.14), and it is used as a model feature.

Incidents cluster in the morning. Reports rise sharply from 6 am and stay at roughly 120 to 140 per hour around 7 to 11 am, dip around 1 pm and rise again around 3 pm. The model predicts daily totals, but this pattern shows that the morning rush is where deployment matters most.

Recent activity is a weak signal on its own. The 7-day average (0.44) and the count 7 days earlier (0.31) correlate positively with today's count, but the scatter plot shows that this mostly reflects which road it is: C5 points sit at higher averages (about 1 to 3.4), while Ortigas and Marcos cluster below about 1. The lag features therefore help only a little beyond the road effect.

Month shows almost no relationship with the target (correlation 0.01), so seasonality is weak in this 18-month period.

Chapter 5 – Machine Learning Model Development

Three regression models were trained on 1,200 road-days (27 Aug 2018 to 30 Sep 2019) and compared with two baselines that predict each road's average (mean) or median count. Linear Regression had the lowest cross-validation RMSE and was selected as the main model.

5.1 Why regression, and why these algorithms

The target is a count (0, 1, 2, ...), so the task is regression, not classification or clustering.

Model	Why it is appropriate
Baseline, mean (average per road)	The simplest sensible forecast of the expected count; any real model must beat it
Baseline, median (median per road)	The best constant guess when error is measured by MAE, because about half of the road-days are zero; the strictest MAE comparison
Linear Regression	Simple and easy to interpret; can predict negative counts, so predictions are clipped at 0
Poisson Regression	Designed for non-negative counts, so it matches the nature of the target
Random Forest Regressor	Captures non-linear patterns and interactions without assuming a formula

5.2 Features and target

Role	Variable	Preparation
Feature	MajorRoad (C5, Ortigas, Marcos)	One-hot encoded
Feature	DayOfWeek (0 to 6)	One-hot encoded
Feature	Month (1 to 12)	One-hot encoded
Feature	Incidents_Lag7	Standardized
Feature	Incidents_Roll7	Standardized
Target	Incidents (count that day)	None

5.3 Training and validation strategy

Incidents form a time series, so a random split would let the model see days after the ones it is tested on. The split is by time instead.

Set	Period	Rows	Mean incidents per road-day
Training	27 Aug 2018 to 30 Sep 2019	1,200	0.88
Testing	1 Oct 2019 to 29 Feb 2020	456	0.86

Models were selected with expanding-window time-series cross-validation (4 folds, cut by date so the three roads of one day are never split) inside the training period, using the lowest cross-validated

RMSE, because planners need the expected count and MAE alone favors predicting the median. The test period was used once, at the end.

Model	Cross-validation MAE (average of 4 folds)	Cross-validation RMSE (average of 4 folds)
Baseline (mean)	0.817	1.043
Baseline (median)	0.710	1.143
Linear Regression	0.765	1.016
Poisson Regression	0.779	1.019
Random Forest	0.788	1.034

5.4 Model configuration and assumptions

Linear Regression uses default settings, and its predictions are clipped at 0. Poisson Regression uses `alpha = 0.001` and `max_iter = 1000`. The Random Forest uses 300 trees, `min_samples_leaf = 5` and `random_state = 42`. One-hot encoding and standardization are part of each model's pipeline and are re-fitted on the training data of every fold.

The models assume that past patterns (road and weekday effects, recent activity) continue, that MMDA reports incidents consistently, that the period from March 2020 is excluded, and that days without a record had no reported incident. The model forecasts one day ahead, because its two history features use the actual counts up to the day before.

5.5 Predictions

The first ten test-period predictions from the selected model (Linear Regression) show its behavior: it follows each road's usual level but never predicts a spike.

Date	Major road	Actual incidents	Predicted
2019-10-01	C5	1	1.99
2019-10-01	Marcos	1	0.82
2019-10-01	Ortigas	2	0.93
2019-10-02	C5	1	2.01
2019-10-02	Marcos	1	0.84
2019-10-02	Ortigas	0	0.99
2019-10-03	C5	2	1.77
2019-10-03	Marcos	0	0.64
2019-10-03	Ortigas	2	0.75
2019-10-04	C5	5	1.84

Chapter 6 – Model Evaluation and Interpretation

On the test period (1 Oct 2019 to 29 Feb 2020), all three models beat the per-road mean baseline on every measure, but only modestly: the best model's average error is 0.75 incidents per road-day against 0.80 for the baseline, and it explains about one third of the variation in daily counts. A per-road median baseline reaches a lower MAE (0.69) but a much worse RMSE and R^2, so the comparison depends on the metric (6.2).

6.1 Results on the test period

Model	MAE	RMSE	R^2
Baseline, mean (average per road)	0.799	1.019	0.273
Baseline, median (median per road)	0.689	1.155	0.067
Linear Regression	0.753	0.976	0.333
Poisson Regression	0.758	0.962	0.353
Random Forest	0.777	1.001	0.299

MAE is the average miss in incidents; RMSE penalizes large misses more; R^2 is the share of the variation the model explains. Lower is better for MAE and RMSE, higher for R^2.

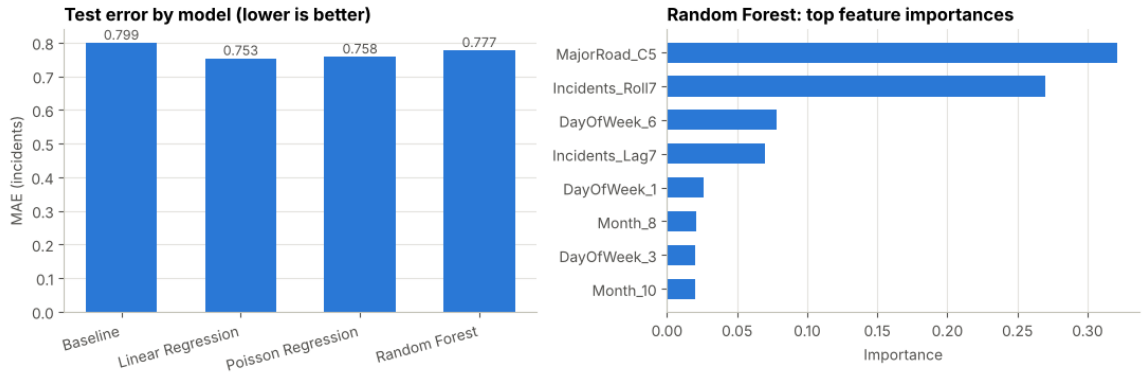


Figure 6.1 – Test error by model (left) and the features the Random Forest relies on most (right).

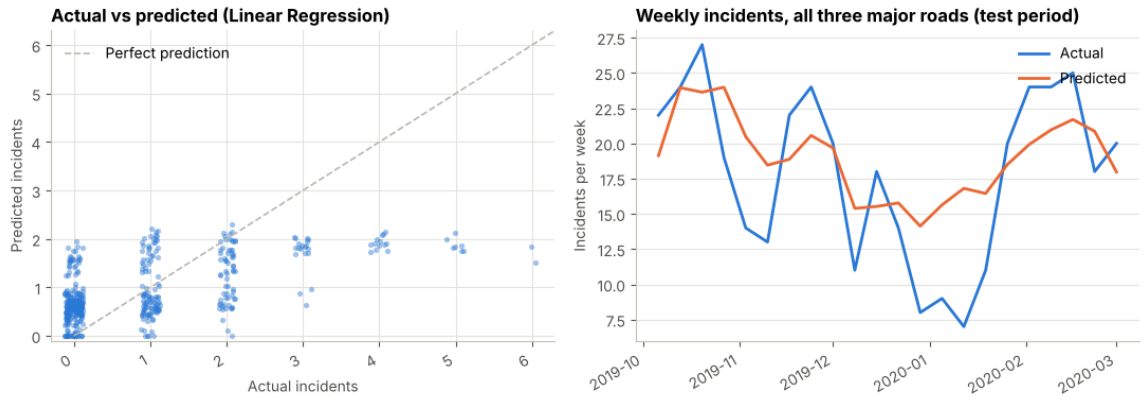


Figure 6.2 – Actual against predicted incidents (left) and weekly totals over the test period (right).

6.2 What the results mean

Linear Regression lowers the MAE by about 6% (0.799 to 0.753), lowers the RMSE from 1.019 to 0.976 and raises R^2 from 0.27 to 0.33, so it is better than the per-road mean baseline, but by a small margin. Poisson Regression has the best RMSE and R^2 (0.962 and 0.353). The ranking depends on the metric: about half of the road-days are zero, and MAE is minimized by predicting the median (0 or 1 for most roads), so the per-road median baseline reaches the lowest MAE (0.689) but the worst RMSE (1.155) and R^2 (0.07), because it never expects a busy day. The models estimate the expected count, which is what planners need, and they are better on RMSE and R^2. The three models are close: cross-validation chose Linear Regression, but the differences are small enough that the choice does not change the conclusions.

The more complex Random Forest did not help and was weaker than the two linear models, which is plausible with only about 1,200 training rows. Most of the signal is simple: the Random Forest relies mainly on whether the road is C5 and on the previous 7-day average, followed by Sunday, which matches the patterns found in Chapter 4.

6.3 Where the model is weak

Major road	Test MAE (Linear Regression)
C5	1.137
Ortigas	0.588
Marcos Highway	0.533

The model never predicts more than about 2 incidents in a day, so it misses the busiest days, which have 4 to 7 incidents. It also misses sudden drops: in the quiet weeks around Christmas and New Year, actual counts fell to about 7 to 9 per week while the model predicted about 15. The error is largest on C5, which has the highest and most variable counts.

6.4 Is the model useful for the intended application?

Partly. It reliably tells planners which road is busiest and that weekdays are busier than Sundays, but that is much of what the per-road average already says, and it does not beat a per-road median on MAE. It does not predict individual busy days well. It is useful as a rough planning guide for where to focus resources, not as a precise forecast of how many incidents will occur on a given day.

6.5 Demonstration: predicting a road and a date

The notebook includes a demonstration cell in which the group picks a major road and a date in the test period, and the selected model (Linear Regression) returns its predicted count next to the baseline and the actual MMDA-reported count, followed by the next 7 days for the same road. For example, for C5 on Tuesday 15 October 2019 the model predicted 2.09 incidents, the baseline 1.60, and MMDA reported 4. Over the following week the predictions stayed between about 1.5 and 2.2 per day while actual counts ranged from 2 to 5, which shows the same behavior described in 6.3: the model follows the usual level but does not predict spikes.

Chapter 7 – Recommendations and Responsible Data Use

The model is not ready for practical use. It is a reasonable starting point for planning, but it needs more data, validation with the MMDA and Pasig City's traffic management office, and a clear human decision-maker before anything depends on it.

7.1 How the results could help stakeholders

C5 should be the first priority when placing enforcers and tow trucks, because it averages 1.64 incidents per day, about three times Ortigas (0.54) and Marcos Highway (0.43). Incidents are lower on Sundays and highest around Tuesday on C5, so weekly schedules can put more units on weekdays. Reports are also highest from about 6 to 11 am, so morning coverage is where extra units matter most.

The model's predicted count per road can serve as a quick baseline expectation of how busy a day is likely to be, to be adjusted by staff who know about events and road works.

7.2 Additional data and validation needed

The model predicts one day ahead, because its features use the previous days' actual counts. To plan a week ahead, as the traffic office would when scheduling, the features would have to be rebuilt with a longer lead (for example, using only information at least 7 days old) and the model re-tested. More data is also needed: incident records beyond what MMDA tweets, the 187 rows without a city, and a longer period that includes 2021 onward, so the model can be tested after the pandemic period. More features would also help, such as holidays and school calendars (the model missed the quiet holiday weeks), rainfall and weather, traffic volume, and scheduled road works or events.

For validation, the predictions should be compared with MMDA's and the city's own incident records and tested on later periods. The stakeholders should be asked what level of error is acceptable, and the model should be tried in a small pilot before anyone relies on it.

7.3 Limitations

Dataset

The dataset holds only incidents that MMDA chose to tweet, and its author notes that it does not represent all incidents in Metro Manila. Because MMDA's jurisdiction is mainly the national roads, the data is biased toward roads such as EDSA and C5, and roads and times that MMDA monitors more closely look busier. The tweets were parsed automatically by a script, so some fields, such as times and locations, contain parsing errors. Many rows have no city, including many on Marcos Highway, so that road's counts are probably too low. Only about 18 months before March 2020 are used, because the author reports a collection gap in early 2020 and traffic also fell during the lockdown. Seventy-four incidents on other roads were left out. Days with no record were treated as days with no incident, which may not always be true, and the coordinates are location-level (about 100 distinct points), not exact positions.

Model

The model explains only about one third of the variation in daily counts (R^2 about 0.33), and it never predicts more than about 2 incidents in a day, so it misses the busiest days. It is a one-day-ahead model, and it does not beat a per-road median baseline on MAE (0.753 against 0.689), only on RMSE and R^2. It does not give an estimate of its own uncertainty, and because it was trained on one period it may not hold when traffic patterns change.

7.4 Data privacy, consent, fairness and responsible AI

On privacy and consent, the data is public information from a government agency and contains no names, plate numbers or other personal details, only vehicle types and places. No individual consent was needed, and the source is cited.

On fairness, a road with few reported incidents is not necessarily safer, because reporting depends on where and when MMDA posts. The results should not be used to blame drivers, neighborhoods or particular vehicle types.

On transparency and human oversight, the model's inputs, weaknesses and errors are documented here and, in the notebook, so decision-makers can judge how far to trust it. Any use should keep a person responsible for deciding where to deploy resources; the model informs the decision and does not make it.

7.5 Is the model ready for practical use?

No. The improvement over a simple per-road average is modest (MAE 0.753 against 0.799, RMSE 0.976 against 1.019), it does not beat a per-road median on MAE (0.689), the model misses busy days and holiday dips, and it was tested on only five months. It needs the additional data and validation in 7.2, and a pilot with the traffic office, before it should support real deployment decisions.

References

Data

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Problem validation sources

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Chua, P. (2020, February 26). For the third year running, C5 is Metro Manila's deadliest road. *Top Gear Philippines*. https://www.topgear.com.ph/news/motoring-news/deadliest-road-c5-metro-manila-sa3557-20200226-src-esquireph (originally published in Esquire Philippines; figures from the MMDA's 2019 MMARAS report).

Top Gear Philippines. (2020, February 24). MMDA reports an all-time high of 121,771 road accidents in 2019. https://www.topgear.com.ph/news/motoring-news/mmda-mmaras-2019-report-a4354-20200224

Tools and libraries (versions used when the notebook was run)

The notebook was run with Python 3.13.17, pandas 3.0.5, NumPy 2.5.3, scikit-learn 1.9.1 and Matplotlib 3.11.2.